

Chapter 12: High-Yield NBDHE Study Guide

Biochemistry, Nutrition, and Nutritional Counseling (English)

Biochemistry, Nutrition, and Nutritional Counseling

1. Carbohydrate Chemistry & Metabolism

- **Monosaccharides:** Simple sugars (Glucose, Fructose, Galactose). Fructose is the sweetest.
- **Disaccharides:**
 - **Sucrose:** Glucose + Fructose (most cariogenic carbohydrate).
 - **Lactose:** Glucose + Galactose (milk sugar).
 - **Maltose:** Glucose + Glucose.
- **Polysaccharides:**
 - **Homopolysaccharides:** Composed of identical repeating monosaccharide units (e.g., Cellulose, Glycogen, Starch).
 - **Heteropolysaccharides:** Mixed carbohydrate units (e.g., Pectin, Hemicellulose, Glycoproteins).
- **Soluble vs. Insoluble Fiber:**
 - **Soluble Fiber:** Pectin, oat bran, gums, mucilages (dissolve in water, lower serum cholesterol).
 - **Insoluble Fiber:** Cellulose, hemicellulose, lignin (increase fecal bulk, accelerate transit).
- **Digestion & Glycolysis:**
 - Salivary amylase begins starch breakdown in the oral cavity.
 - In aerobic glycolysis, glucose produces **Pyruvate**, which converts to **Acetyl-CoA** for the Krebs cycle.
 - In anaerobic glycolysis, glucose produces **Lactic acid**.
- **Cariogenicity & Stephan Curve:**
 - Frequency and form (retentiveness/stickiness) of sugar intake are far more critical to caries risk than total quantity consumed.
 - Each sugar exposure drops plaque pH below critical level (**5.5 for enamel**, **6.0–6.7 for cementum/dentin**) for 20–40 minutes.
 - Eating sweets with meals rather than between meals reduces net cariogenic exposure.

2. Amino Acids & Protein Metabolism

- **Essential vs. Non-Essential Amino Acids:**
 - Essential (must be supplied in diet): Histidine, Isoleucine, Leucine, Lysine, Methionine, Phenylalanine, Threonine, Tryptophan, Valine.
 - PKU (Phenylketonuria): Patients lack phenylalanine hydroxylase; avoid **Aspartame** (NutraSweet) which hydrolyzes to phenylalanine.
- **Digestion:** Pepsin (stomach) and trypsin/chymotrypsin (pancreas) hydrolyze peptide bonds.
- **Nitrogen Balance:**
 - **Positive Balance:** Intake > Output (Growth, pregnancy, recovery from trauma).
 - **Negative Balance:** Output > Intake (Starvation, catabolic illness, anorexia nervosa).

3. Lipids & Energy Metabolism

- **Caloric Values:**
- **Lipids: 9 kcal/g**
- **Carbohydrates: 4 kcal/g**
- **Proteins: 4 kcal/g**
- **Alcohol: 7 kcal/g**
- **Compound Lipids:** Glycolipids (contain carbohydrate components and are abundant in brain and myelin sheath).
- **Bile Salts:** Synthesized by liver, stored in gallbladder; emulsify dietary fats in duodenum.

4. Vitamins & Minerals in Dental Health

- **Fat-Soluble (A, D, E, K):**
- Stored in liver and adipose tissue; toxicity possible with chronic mega-doses.
- **Vitamin A:** Essential for salivary gland epithelium integrity, amelogenesis, and mucosal maintenance.
- **Vitamin D:** Facilitates calcium absorption; deficiency causes rickets (children) or osteomalacia (adults).
- **Vitamin E:** Antioxidant membrane protection.
- **Vitamin K:** Co-factor for prothrombin and clotting factors (II, VII, IX, X).
- **Water-Soluble (B-Complex, C):**
- Readily excreted in urine; daily intake required.
- **Vitamin C (Ascorbic Acid):** Essential cofactor for prolyl/lysyl hydroxylase in collagen synthesis (dentin, pulp, PDL). Deficiency causes Scurvy.
- **Thiamin (B1) & Niacin (B3):** Depleted by chronic alcoholism.
- **Vitamin B12 (Cobalamin):** Bound to intrinsic factor in stomach; deficiency causes ****Pernicious Anemia**** (glossitis, red beefy tongue, paresthesia). Vegans require supplementation.

5. Clinical Nutritional Counseling & Special Populations

- **24-Hour Recall + 3–7 Day Food Diary:** Golden standard for assessing accurate dietary habits.
- **Diabetic Patients:** Schedule morning visits 1.5–2 hours post-breakfast and medication to prevent hypoglycemia.
- **CKD (Chronic Kidney Disease):** Restrict dietary phosphorus, sodium, and carefully titrate protein.
- **New Prosthesis:** Soft/liquid diet for first 24 hours; chew bilaterally to equalize occlusal load; flavors initially feel altered.
- **Bulimia & Anorexia:** Perimylolysis (chemical acid erosion of lingual maxillary surfaces) and parotid gland hypertrophy.